# 1. Requirements definition

The requirement is to make all the calculation features of the official Qalculate! usable on an Android phone (arm64) as an ordinary app, "QCalc", that anyone can install from Google Play.

## 1.1 Purpose and background

The purpose is to make all the calculation features of the official Qalculate! usable on an Android phone (arm64) and to let anyone install them from Google Play. The target phone is the SHARP AQUOS sense8 (Android 16, arm64, a 1080 × 2432 screen).

- The official Qalculate! runs on computers (the GTK and Qt versions, and the command-line qalc). There is no official Android version.
- The user asked for an app that installs like any Android app, is built from the latest official source code, and is published on Google Play.
- So the official calculation library, libqalculate 5.13.2, is built in without changes, and only the user interface is new, made for phones. Expressions are written and results come out as in the official program (QCalc adds only 10 traditional Japanese units and trace()).
- Version 1.0.0 took the name QCalc and met the conditions of Google Play (an App Bundle, targetSdk 36, a privacy policy, the license texts). At the user's request it also added 3D graphs (z = f(x, y)).
- Version 1.1.0, asked to be "easier to use" and to "handle calculus and matrices", got four pages of keys (123, Functions, Calculus, Matrix), matrix results shown as tables, examples, undo, text sizes, search in Japanese and the traditional Japanese units.
- Version 1.2.0 raised the highest precision from 1000 to 200,000 digits, the most Android handles without trouble.
- Version 1.3.0 made the app English and Japanese, with English as the default ("Language / 言語" switches), added seven color themes and exporting the history as a CSV file, shows all digits of long results in the preview too (the preview of 4587! had shown only its beginning, without the 1,144 zeros at the end that the history showed), fixed digit grouping, and prepared QCalc Pro, a one-time purchase in Google Play (1.9).
- Version 1.4.0 reviewed the whole app. It suggests entries below the input field (after "to", the units a value can be converted to; while typing with the phone's keyboard, the names of functions, units and constants), adds a setting for a light or dark display, and shows the date and time of a calculation in the menu of its history item. It also fixes problems found in the review: "to mixed" answered with the Japanese units, a "to" at the end of an expression previewed a meaningless result, and in landscape a long expression pushed the keys off the screen.

## 1.2 Scope

The scope is the calculation features of libqalculate and what a phone needs to use them: the screens, saving and updating.

| | Item |
|---|---|
| In | Arithmetic and functions, units, currencies, equations (solve), derivatives and integrals, matrices, number bases, dates, uncertainty propagation, complex numbers, data sets (elements and others), a precision of up to 200,000 digits |
| In | The on-screen keys (four pages: 123, Functions, Calculus, Matrix) and the phone's keyboard, the result while typing, suggestions (conversion targets and names), matrices as tables, calculating and the history, ans, stopping a calculation |
| In | Lists of functions, variables and units with search, graphs (2D and 3D), user-defined variables and functions, exchange-rate updates, settings, examples and tips, undo for a cleared expression or history |
| In | English and Japanese screens (English by default), automatic conversion of Japanese input (full-width characters, 円, —), a light or dark display (the phone's setting or a choice), color themes, landscape, screen readers (TalkBack) |
| In | Exporting the history as a CSV file |
| In | QCalc Pro: unlocking some features with a one-time purchase in Google Play (only in builds with the Google Play Billing Library; 1.9) |
| In | Distribution on Google Play (App Bundle, store images and texts in English and Japanese, privacy policy), the license texts of the bundled software and publishing the source code |
| In | The traditional Japanese units (tsubo, tan, sun, shaku, ken, ri, gō, shō, monme, kan) and the matrix trace trace(): the only calculation features QCalc adds to libqalculate |

| | Item |
|---|---|
| Out | Phones that are not arm64 (32-bit ARM, x86) |
| Out | Plotting with gnuplot as in the official version; file operations and external commands from expressions, such as command(), load() and export() (the app has its own graphs instead) |
| Out | RPN mode, screens in languages other than English and Japanese, sync between devices |
| Out | Functions that give eigenvalues or eigenvectors directly, differential equations of second or higher order, the values of infinite sums (libqalculate does not have them; eigenvalues can be found by solving the characteristic equation with solve) |

## 1.3 Users and environment

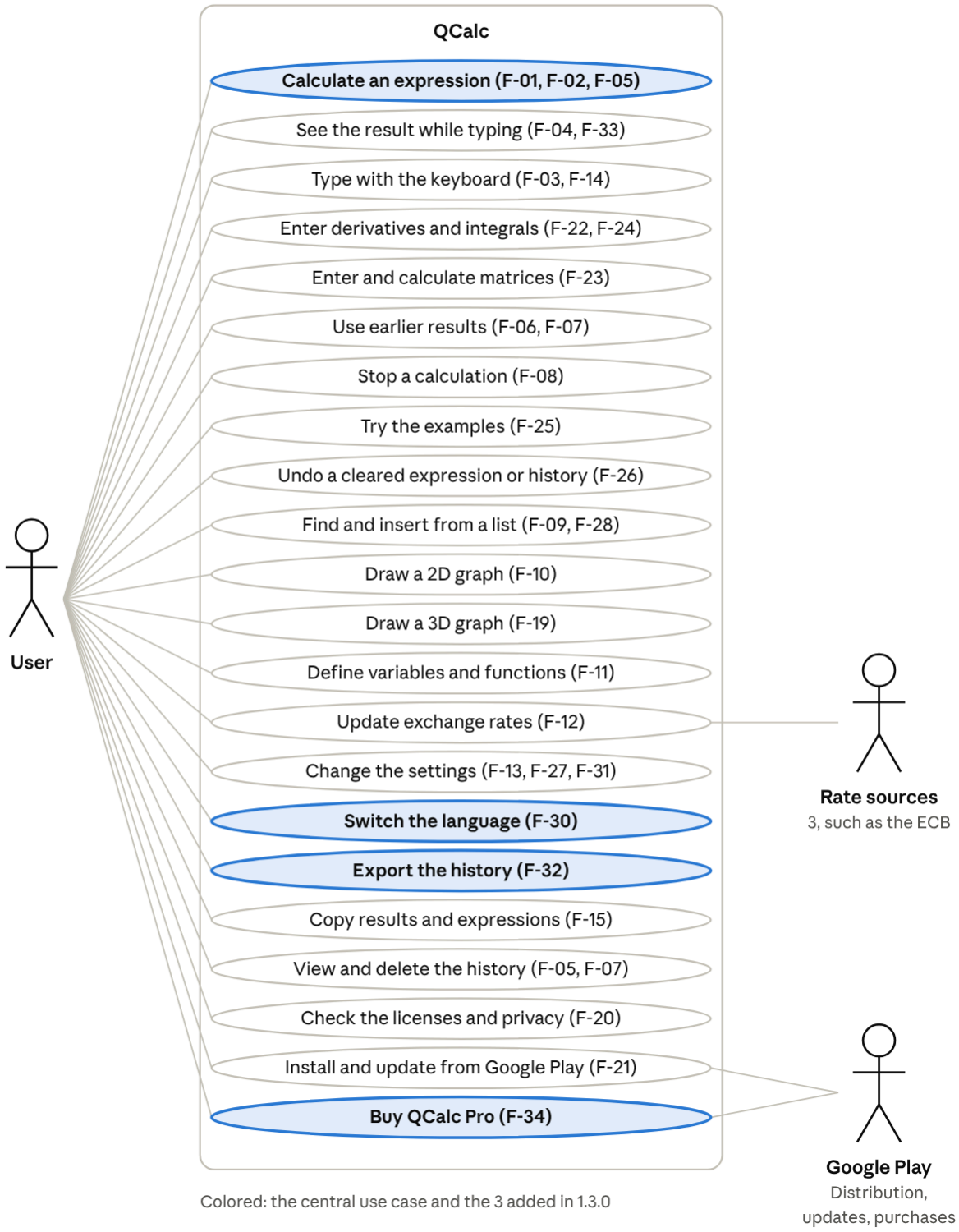
The users are individuals who use English or Japanese. Because QCalc is published on Google Play, anyone can install it on their own phone. Calculations need no network.

| Item | Contents |
|---|---|
| Users | Individuals who read English or Japanese, for technical calculations (units, number bases, many digits) and everyday calculations |
| Supported phones | Android 7.0 (API 24) to 16 (API 36) with an arm64-v8a CPU |
| Target phone | SHARP AQUOS sense8 (Android 16, a 1080 × 2432 screen) |
| Orientation | Mainly portrait. In landscape with a height of 560 px or less, the history and the input are on the left and the keys on the right |
| Network | Not needed for calculations. The app goes online only to update exchange rates (and, in builds with billing, to talk to Google Play). The links in "About this app" (privacy policy, source code, Qalculate! website) open in the phone's browser |

| Item | Contents |
| --- | --- |
| Distribution | Google Play (Google makes APKs for each phone from the App Bundle). During development a test APK with another package name can be installed directly |
| Permissions | Only INTERNET and VIBRATE, both granted without asking the user. Builds with billing also get com.android.vending.BILLING from the Billing Library |

# 1.4 Use cases

**The user's 22 use cases; outside, only exchange rates and Google Play**
In parentheses: the IDs in 1.5 Functional requirements




Use case diagram · 22 use cases

There is one kind of user (whoever uses the app). The only outside systems are the sources of exchange rates and Google Play (distribution, updates and, with billing, purchases). F-16 (dark theme), F-17 (landscape) and F-18 (keeping the state) are not single

actions but apply to every use case.

| Use case | Main flow |
| --- | --- |
| Calculate an expression | Type with the keys and press =. The result stays in the history (stopped after 20 seconds) |
| See the result while typing | When typing pauses, the result appears under the input field. Expressions that change something (assignments) are not run. After "to" (the → key), the units to convert to are suggested and a tap types one (1.4.0) |
| Type with the keyboard or in Japanese | ⌨ switches to the phone's keyboard; Enter calculates. Full-width characters, 円 and ー are converted. Functions, units and constants that begin with the letters typed are suggested (1.4.0) |
| Enter derivatives, integrals and limits | The Calculus page → d/dx, ∫, lim and other keys write templates; write the expression. The "=" of an equation is the a=b key |
| Enter and calculate matrices | The Matrix page → type [1, 2; 3, 4] and calculate det, transpose, inverse, products. Matrix results are tables |
| Use earlier results | Press ans, or tap a result in the history to insert its exact value |
| Stop a calculation | Tap "Calculating… (tap here to stop)" |
| Try the examples | Tap an example on the first screen, or menu → "Examples and tips" → tap one |
| Undo a cleared expression or history | Tap "Undo", shown right after AC, after an expression was replaced, or after deleting or clearing the history |
| Find and insert from a list | Menu → a list → search (also in Japanese, on the Japanese screen) → "Insert" or "Convert" |
| Draw a 2D graph | Menu → Graph → 2D → expressions and range → Draw. Move, zoom, tap for coordinates |
| Draw a 3D graph | Menu → Graph → 3D → expressions in x and y, ranges, grid → Draw. Drag to turn, pinch to zoom, tap for coordinates |
| Define variables and functions | Menu → User-defined → name and value → Save |

| Use case | Main flow |
|---|---|
| Update exchange rates | Menu → Update exchange rates → Update now. Automatically at start when older than 7 days |
| Change the settings | Menu → Settings. Angle, exact/approximate and number base also with the chips at the top. The light or dark display is chosen here too (1.4.0) |
| Switch the language | Menu → "Language / 言語" (or the first setting) → English or 日本語 |
| Export the history | Menu → "Export history (CSV)" → choose where to save |
| Copy results and expressions | Touch and hold a history item → "Copy result", "Copy expression" |
| View and delete the history | Scroll the history. Touch and hold for the menu, which shows when the item was calculated and deletes one item; "Clear history" in the menu clears all |
| Check the licenses and privacy | Menu → About this app → "Privacy policy" (browser), "Open source licenses" (full texts in the app) |
| Install and update from Google Play | Install from the Play Store. New versions come as Play Store updates; the history and settings stay |
| Buy QCalc Pro (builds with billing) | Touch a locked feature or menu → "QCalc Pro" → Buy (Google Play's purchase screen); "Restore a purchase" asks Google Play again |

## 1.5 Functional requirements

There are 35 functional requirements. F-01 to F-33 and F-35 are met in version 1.4.0 (F-22 to F-28 were added in 1.1.0, F-29 in 1.2.0, F-30 to F-34 in 1.3.0, F-35 in 1.4.0). F-34 works only in builds with the Google Play Billing Library (1.9). The right column says how each is met.

| ID | Requirement | How it is met |
|---|---|---|
| F-01 | The calculation features of the official Qalculate! can be used, written the same way | libqalculate 5.13.2 without changes; the expression is passed to it as typed |

| ID | Requirement | How it is met |
|---|---|---|
| F-02 | Expressions can be typed with on-screen keys | Four pages of keys (5 × 5 each: 123, Functions, Calculus, Matrix) switched with tabs, and above them 8 keys for every page (x, comma, a=b, →, ans, ⌨, ◀, ▶) |
| F-03 | The phone's keyboard can be used too | The ⌨ key switches; Enter calculates |
| F-04 | The result is shown while typing | Calculated 160 ms after the last input, stopped after 1.5 seconds. Expressions with assignments are not run. Can be turned off |
| F-05 | = calculates and keeps the result in the history | 20-second limit. Up to 300 history items |
| F-06 | The last result can be used as ans | Kept with the same rules as the official qalc, and restored after the app closes |
| F-07 | Results and expressions in the history can be reused | Tap a result to insert its exact value, tap an expression to edit it, touch and hold for a menu (copy, edit, delete; it also shows the date and time of the calculation) |
| F-08 | A long calculation can be stopped | Tapping "Calculating…" stops the engine; it starts again with the next calculation |
| F-09 | Functions, variables and units (with currencies) can be listed and inserted | Lists from the engine, with categories and search; search puts the best match first and, on the Japanese screen, also works with Japanese names. A unit is inserted in a spelling that the engine has confirmed reads as that unit |
| F-10 | 2D graphs (y = f(x)) | Up to 6 expressions, 600 intervals each. Move, zoom, coordinates |
| F-11 | Variables and functions can be defined and saved | Saved in the engine's definitions file; expressions the engine cannot read are refused |
| F-12 | Exchange rates can be updated | The same three sources as the official version. Manual updates, and automatic ones when the rates are older than 7 days |

| ID | Requirement | How it is met |
|---|---|---|
| F-13 | Calculation and display settings can be changed | 20 settings: language, angle, approximation, fractions, precision (2 to 200,000 digits), number base, exponential notation, light or dark, text size, color theme and more. The three most used also with buttons at the top |
| F-14 | Japanese input is calculated as typed | Full-width letters, digits and signs, 円 and ￥, and ― between digits are converted in the input field |
| F-15 | Results and expressions can be copied | The phone's clipboard |
| F-16 | Dark theme | Follows the phone's setting. The setting "Light or dark" can also keep it always light or always dark (1.4.0) |
| F-17 | Landscape | A landscape layout |
| F-18 | The history, settings, the expression being typed and ans are kept | Saved at once to files and SharedPreferences, restored after a restart |
| F-19 | 3D graphs (surfaces z = f(x, y)) | Up to 4 expressions; grids of 25 × 25, 41 × 41 or 61 × 61. The engine calculates the grid points, the page draws the surface in 3D. Drag to turn, pinch to zoom, tap for coordinates. The z range is automatic (extreme values left out). 15-second limit |
| F-20 | The privacy policy and the open source licenses can be read | "About this app" opens the privacy policy (GitHub Pages) and the source code, and shows the full license texts of the app and of the 7 bundled pieces of software (8 in all) in the app |
| F-21 | Installed from Google Play, with updates | The App Bundle is signed with the upload key and distributed with Play App Signing. Updates raise VERSION_CODE |
| F-22 | Derivatives, integrals, limits, sums and differential equations can be typed with on-screen keys | The Calculus page. Keys write templates such as diff(, x, 2), integrate(, 0, 1), limit(, 0), sum(, 1, 10) and dsolve(diff(y, x) = ) and put the cursor where the expression goes. A selection is wrapped in the template. Keys for ∞ and y too |

| ID | Requirement | How it is met |
|---|---|---|
| F-23 | Matrices can be typed with on-screen keys and their results seen as tables | The Matrix page (digits, [ ] ;, det, transpose, inverse, identity, × − +). Results with two or more rows are tables in the history and the preview, with smaller characters when too wide |
| F-24 | The "=" of equations and the lists of functions by field are one key away | a=b writes " = " (the big = key calculates). "more…" on the Calculus and Matrix pages opens the function list at that category |
| F-25 | Examples and tips can be read and tried in the app | "Examples and tips" in the menu: 69 expressions in 12 groups and 12 tips. The first screen (no history yet) shows how to start and 4 examples. Tapping an example puts it into the input field |
| F-26 | A cleared expression or history can be brought back | After AC or an expression replaced from the history or the examples, "Undo" appears at the top right of the input field (until something is typed). A deleted history item can be brought back for 5 seconds, a cleared history for 6 seconds |
| F-27 | The text size can be chosen | "Text size": Normal, Large (1.18 ×), Extra large (1.38 ×), for the history, the input field, the result and the messages |
| F-28 | Traditional Japanese units and the matrix trace | At start, the bridge program adds tsubo, tan, sun, shaku, ken, ri, gō, shō, monme and kan (category "Japanese Units") and trace() to the engine. They are not saved in the user's definitions file |
| F-29 | Long results (200,000 digits and the like) are shown and copied with all digits, and kept in the history and ans | All digits are shown; a long result scrolls in a box (up to 70 % of the history's height). Copying copies all digits. The history keeps the newest results in full up to 1,000,000 characters; older long results keep their first 2,000 characters. ans keeps values up to 500,000 characters and restores all digits after the engine restarts |

| ID | Requirement | How it is met |
|---|---|---|
| F-30 | The app is in English and Japanese | English by default. "Language / 言語" in the menu and in the settings switches; the page loads again in that language with the history and the expression kept. The page carries both languages; the Java code and the engine report in English and the page translates their messages into Japanese |
| F-31 | Color themes | Seven themes (blue, green, teal, purple, rose, orange, graphite) for the keys and highlights, each in the light and the dark theme, with a contrast of 4.5:1 or more |
| F-32 | The history can be exported | A CSV file (UTF-8 with a byte order mark): date and time, expression, result as shown, and the value as a plain number when there is one. Saved where the user chooses with Android's file picker (no storage permission) |
| F-33 | The preview shows the same as the history | The preview gets results as long as the history (450,000 bytes). A longer result is shown, in both, as its beginning, "…" and its last 20,000 bytes, with a note giving the number of digits |
| F-34 | QCalc Pro can be bought once and unlocks Pro features | The one-time product qcalc_pro in Google Play. Pro unlocks a precision above 1000 digits, 3D graphs, color themes and the CSV export (and later Pro features). Builds without the Billing Library lock nothing; users from before Pro keep the Pro features |

| ID | Requirement | How it is met |
|---|---|---|
| F-35 | Entries can be chosen from suggestions | Suggestions appear below the input field. After "to" (the → key): the units of the same kind as the unit in the expression, the most used first (without a unit: ways to write a number, such as a fraction or hexadecimal). While typing with the phone's keyboard: the functions, variables and units whose names begin with the letters typed (on the Japanese screen, Japanese names find them too). A tap types the entry; a function gets its parentheses with the cursor inside |

## 1.6 Non-functional requirements

At the center: "Whatever is typed, the app never freezes, and the result shown never differs from the value kept."

| Area | Requirement | How it is met |
|---|---|---|
| Performance | The preview stops after 1.5 s, a calculation after 20 s, a 2D graph after 12 s, a 3D graph after 15 s | Every request carries its time limit, measured in libqalculate's calculation thread. The engine starts together with the screen |
| Performance | At the highest precision (200,000 digits), π, e, √2, sin(1), ln(2) and the like finish within the time limit | MPFR as it is (0.3 to 1.2 s on one PC core). Heavy functions such as ζ(3), which do not finish at this precision, are stopped at the limit. A long saved ans is read directly with GMP (about 0.2 s for 200,000 digits) |
| Performance | Long results do not make typing slow | Only one preview runs at a time, after typing pauses; a result that arrives after the expression changed is not shown, and the next preview starts |

| Area | Requirement | How it is met |
|---|---|---|
| Reliability | The app does not freeze when a calculation does not end | A watchdog in the engine returns a timeout at the limit + 3 s and exits; the Java side kills the engine at the limit + 10 s and starts it again with the next request |
| Reliability | The app recovers when the engine crashes; the request that crashed it is not sent again | A request is sent again only when writing it failed. If the page's renderer stops, only the page is made again |
| Correctness | The result shown, ans and values taken from the history agree | An expression is evaluated once and that value is kept as ans. Insertion uses the exact text the engine returns |
| Correctness | Results are the same as the official qalc's | The expression goes to libqalculate as typed; even the conditions for showing exact forms match qalc. Checked with 317 expressions |
| Correctness | No warning or error is lost | libqalculate 5.13.2 leaves a message hold open after a definite integral with an infinite bound, hiding later warnings and errors (the official qalc too). The bridge program closes it for every command |
| Data safety | A forced stop or power loss does not lose the history or definitions | Written to a temporary file, fsync, then renamed. A damaged definitions file is moved aside, never overwritten |
| Security | Nothing outside can make the app show arbitrary pages or run commands | The WebView shows only the app's own page, with file access off. External links open in the browser, http and https only. The engine's command() and the like are disabled. The file the page asks to save must be a CSV with a plain name |

| Area | Requirement | How it is met |
|---|---|---|
| Privacy | Calculations never leave the phone | The only network use is downloading exchange rates (3 sources, download only), and with billing, Google Play. No personal data is collected. The privacy policy is published; Google Play's data safety says "no data collected" |
| Usability | Easy to use in English or Japanese | Screens in both languages, Japanese translations of the common messages of the engine and the Java code (57 kinds), a layout that keeps the keys visible with long results. The first screen shows how to start and examples (all visible at 360 × 740); every page of keys is one tab away |
| Usability | Works with a screen reader (TalkBack) | Keys have names, ordinary clicks work, only the result of a calculation is announced |
| Compatibility | Runs on Android 7.0 to 16 | targetSdk 36. Edge-to-edge drawing on Android 15 and later and the back gesture of Android 13 and later are handled. A static arm64 executable aligned for 16 KB pages |
| Maintainability | Easy to update to new official versions | libqalculate unchanged. Builds are reproduced with shell scripts (fetch-sources.sh, build-engine.sh, build-apk.sh, build-aab.sh) |
| License | Follows libqalculate's GPL v2 or later | The app's code is GPL v2 or later too, with a permission to link the Google Play Billing Library. The sources, including the bundled libraries, are on GitHub; the license texts are in the app |

| Area | Requirement | How it is met |
| --- | --- | --- |
| Distribution | Meets Google Play's conditions (App Bundle, targetSdk 36 or later, 64-bit, 16 KB pages, privacy policy, for purchases, a supported version of the Billing Library) | build-aab.sh makes the App Bundle and bundletool 1.17.2 checks it. Store images and texts are in store/, the privacy policy in docs/. The billing code uses Billing Library 8.3.0 |

## 1.7 Constraints and assumptions

Much of the design follows from one constraint: do not touch the official calculation code.

1. libqalculate's source is not changed. What is specific to the app lives in a bridge program that calls libqalculate's public functions, including the Japanese units, trace() and closing the message hold.
2. The engine is built with zig (a cross compiler), without the Android NDK. The test APK and the App Bundle are built with Ubuntu's Android tools (aapt2, dx, zipalign, apksigner, jarsigner) and the Android SDK's android.jar (API 36 and 34) (build-apk.sh, build-aab.sh). The build environment cannot reach Google's servers, so the App Bundle is assembled without bundletool, which is only used to check it. The same sources open in Android Studio (Gradle), but that build has not been tried.
3. The engine is a Linux executable stored in lib/arm64-v8a/ under a library name. Since Android 10, files in an app's data cannot be executed, so the engine starts from the native library directory, where it is extracted at installation (extractNativeLibs=true). The App Bundle keeps native libraries compressed in the APKs Google Play makes (BundleConfig: uncompressNativeLibraries off).
4. targetSdk is 36 (Google Play's condition for new apps), so the app handles edge-to-edge drawing on Android 15 and later and the back gesture changed in Android 16.
5. Distribution is on Google Play, and Google keeps the app signing key (Play App Signing). The local upload key is never published. Earlier builds (jp.local.qalculate) are a different app and cannot be updated in place.
6. The name and the icon differ from Qalculate!'s, and the description says "not an official Qalculate! app", so that QCalc is not taken for the official app (F-Droid also has another unofficial app with the same name).
7. A new personal developer account must run a closed test with at least 12 testers for 14 days before applying for production.
8. The engine's messages and the names in its lists are in English (libqalculate has no Japanese translation). In Japanese, the page translates the common messages and the category names, and shows Japanese names beside functions, units and

constants (their descriptions stay in English).

9. The phone's text size setting is not followed (the zoom is fixed at 100 % to keep the keys in place). The app's own setting offers three text sizes instead.
10. Exchange rates depend on the public data of their sources. When they cannot be downloaded, the built-in or last downloaded rates are used.
11. The highest precision is 200,000 digits. Android limits one transfer between an app and the system to about 1 MB (Binder), which includes copying to the clipboard. A 200,000-digit real result, even with digit grouping, is about 270,000 characters (about 530,000 bytes in UTF-16) and can be copied. A higher precision would give results that cannot be copied, and sin, ln and the like would come close to the 20-second limit.
12. English is the default at every start until a language is chosen, also on phones set to Japanese and for users updating from 1.2.0, as the user asked. The release notes say how to switch.
13. The purchase of QCalc Pro needs the Google Play Billing Library (8.3.0) and about 20 libraries it uses, from Google's Maven repository, which the build environment cannot reach. So the shell builds leave billing out (and lock nothing), and billing is built with Gradle (qcalc.billing=true) where Google's repository is reachable.

Limits of the engine itself (libqalculate 5.13.2), which QCalc does not work around (the user manual 4.4 and 4.5 tell users):

- Every limit carries the note that it is "incomplete and unreliable".
- Only first-order differential equations are solved. Infinite sums (Σ up to ∞) get no value.
- No function gives eigenvalues or eigenvectors (solving the characteristic equation with solve works).
- Single-letter names (a, b, m, t and others) are read as units; use x, y and z as variables.

## 1.8 Acceptance criteria

18 of the 22 criteria have been checked on a PC (A-15 to A-18 were added in 1.1.0, A-19 in 1.2.0, A-20 and A-21 in 1.3.0, A-22 in 1.4.0). The 4 that need a phone (A-02, A-10, A-11, A-13) are checked in Google Play's internal test.

| ID | Criterion | How checked | Status |
|---|---|---|---|
| A-01 | The App Bundle and the test APK are signed and built correctly | bundletool 1.17.2 (validate, build-apks), jarsigner, apksigner, aapt2 (22 checks) | Checked on a PC |

| ID | Criterion | How checked | Status |
|---|---|---|---|
| A-02 | Installs and starts on a phone (Android 7.0 or later, arm64) | Installed on the target phone from Google Play's internal test | Not yet |
| A-03 | The typical expressions of the test specification give the same results as the official qalc | The arm64 engine run on a PC under emulation and compared (317 expressions) | Checked on a PC |
| A-04 | The preview does not run assignments (x = 5 and the like) | Engine tests and automated UI tests | Checked on a PC |
| A-05 | The result and ans agree, and are restored after the app closes | Engine tests and tests running the Java code on a PC | Checked on a PC |
| A-06 | A calculation that does not end does not freeze the app, which recovers by itself | fibonacci(10^9) in exact mode (times out) and an engine made not to answer | Checked on a PC |
| A-07 | Reusing a history result does not change its value | Automated UI tests | Checked on a PC |
| A-08 | Input with full-width characters, 円 and ー is calculated correctly | Automated UI tests (not yet with a phone's Japanese keyboard) | Checked on a PC |
| A-09 | Google Play's technical conditions are met | targetSdk 36, App Bundle, arm64 only, 16 KB alignment (readelf), signed with the upload key | Checked on a PC |
| A-10 | Usable with TalkBack | On a phone with TalkBack on | Not yet |
| A-11 | Exchange rates are really downloaded and updated | "Update now" on a phone | Not yet |
| A-12 | 3D graphs are drawn, turned and zoomed, and show coordinates; expressions without values and wrong ranges are reported correctly | Automated UI tests (with the real arm64 engine) | Checked on a PC |
| A-13 | On Android 15 and later the page is not hidden by the system bars, and the back gesture closes panels | On a phone (Android 16) | Not yet |
| A-14 | The privacy policy and the licenses can be read from the app | Automated UI tests. The privacy policy (GitHub Pages) was confirmed online on 2026-11-12 | Checked on a PC |

| ID | Criterion | How checked | Status |
|---|---|---|---|
| A-15 | Expressions can be typed with the Calculus and Matrix keys; templates, wrapping a selection, ⌫, a=b and "more…" work | Automated UI tests (W01 to W13) | Checked on a PC |
| A-16 | Matrix results are tables, and inserted text reads as the same matrix | Automated UI tests (W11 to W15, W21) and engine tests (M10 to M18) | Checked on a PC |
| A-17 | All examples in the app can be calculated; undo, text sizes and search in Japanese work | The 73 example expressions run by the engine; automated UI tests (V01 to V21) | Checked on a PC |
| A-18 | Warnings and errors are shown after a definite integral with an infinite bound | Engine tests (M04, M05) and automated UI tests (W17) | Checked on a PC |
| A-19 | At a precision of 200,000, all digits of π and the like are shown and copied correctly, and the latest results and ans keep all their digits after the app opens again | Engine tests (H01 to H16, with a full comparison against the mpmath library), Java tests (J15 to J21), automated UI tests (L01 to L32) | Checked on a PC |
| A-20 | English by default; switching to Japanese and back keeps the history; the preview equals the history for long results; digit grouping puts commas only before the decimal point | Automated UI tests (Q01 to Q03, Q20 to Q27) and the earlier UI tests run in Japanese (W, V, fixes, 3D) | Checked on a PC |
| A-21 | Pro features are locked only when Pro can be bought and is not owned; buying unlocks them; color themes and the CSV export work | Automated UI tests with the test server's billing modes (Q04 to Q19) | Checked on a PC |

| ID | Criterion | How checked | Status |
|---|---|---|---|
| A-22 | Names and conversion targets can be entered from the suggestions below the input field; the setting "Light or dark" overrides the phone's setting; the history menu shows the date and time; "to mixed", a "to" at the end of an expression and a long expression in landscape are handled correctly | Automated UI tests (T01 to T27) and 8 "to mixed" expressions added to the comparison with qalc | Checked on a PC |

## 1.9 QCalc Pro

QCalc Pro is a one-time purchase (the one-time product qcalc_pro) that unlocks:

| Feature | Free | With Pro |
|---|---|---|
| Precision (significant digits) | Up to 1,000 | Up to 200,000 |
| 3D graphs | — | Yes |
| Color themes | Blue | Seven colors |
| Export the history (CSV) | — | Yes |
| Pro features of later versions | — | Yes |

- Pro is unlocked by a purchase in Google Play (with the Google Play Billing Library), and in test builds (package name ending in .debug), where the Pro panel can lock it for testing.
- A build without the Billing Library cannot sell Pro, so it locks nothing: everything is free, as before Pro. The 1.3.0 and 1.4.0 App Bundles made by the shell scripts are such builds.
- Users who used QCalc before Pro began (their phone has a saved history or settings, or ran a build without billing) keep the Pro features for free, because 1.2.0 already gave them high precision and 3D graphs.
- The purchase is acknowledged after payment; a pending payment unlocks Pro when it completes; purchases are asked for at every start and return to the app, so purchases made elsewhere (promo codes) and refunds are noticed.